Addition & Subtraction Patterns on the Open Number Line

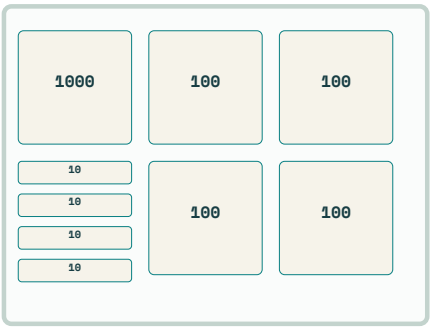
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.NBT.A.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Place Value Base Ten

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
To add 48 + 35, shift 2 unit cubes to 48 to make 5 tens rods (50), leaving 33.

Sketch Model on Paper:
Draw an open number line showing friendly jump to benchmark 50.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	50	X: 50	Benchmark 50 (48+2)
2.	ENTER	Y: 50 | X: 50	Stack lift
3.	33	Y: 50 | X: 33	Compensated 33 (35-2
4.	+	X: 83	Sum: 50 + 33 = 83

EXPECTED LCD X: 83

1 **Solve 49 + 37 using compensation (50 + 36).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Solve 68 + 25 using compensation (70 + 23).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Solve 195 + 150 using compensation (200 + 145).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Solve 298 + 124 using compensation (300 + 122).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 1

Addition & Subtraction Patterns on the Open Number Line • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Place Value Base-Ten Studio (BASE-10)

CAD Blueprint: place_value_base_ten

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/place_value_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Place Value Tray & Base-Ten Blocks (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 49 + 37 using compensation (50 + 36).** Standard: CCSS.MATH.3.NBT.A.2	86	[5][0] [ENTER] [3][6] [+]
2	**Solve 68 + 25 using compensation (70 + 23).** Standard: CCSS.MATH.3.NBT.A.2	93	[7][0] [ENTER] [2][3] [+]
3	**Solve 195 + 150 using compensation (200 + 14** Standard: CCSS.MATH.3.NBT.A.2	345	[2][0][0] [ENTER] [1][4][5] [+]
4	**Solve 298 + 124 using compensation (300 + 12** Standard: CCSS.MATH.3.NBT.A.2	422	[3][0][0] [ENTER] [1][2][2] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for forgetting to subtract the compensation amount added in step 1.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.